This Assignment will not be considered for Course participation Marks after 16th Oct 2022

Instructions for Extended Match Questions

Use this slide to understand how to answer an EMQ

Instructions:
1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. Please click here to view a sample template file for answering.
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

List of Options:
1.Linearly independent set.
2.Linearly dependent set.
3.If we remove any one of the vectors from the set, then it will be linearly independent.
4.If we put the vector $(1, 1, 1)$ in the set, then it will be linearly dependent.
5.There exist one vector in the set, if we remove that vector from the set, then it will be linearly independent.
6.If we remove any two vectors from the set, then the set will be linearly independent.

List of Questions:

Q1. Consider the set $S_1 = \{(1, 0, 1), (1, 0, 0), (1, 1, 0), (0, 1, 0)\}$. Which of the above options is(are) true for S_1?

Q2. Consider the set $S_2 = \{(1, 1, 0), (1, 1, 0) + (0, 1, 1), (1, 1, 0) - (0, 1, 0)\}$ with usual coordinate-wise addition and scalar multiplication. Which of the above options is(are) true for S_2?

Q3. Consider the set of all points on X-axis, each one treated as a different vectors in $\mathbb{R}^3$, with usual addition and scalar multiplication. Which of the above options is(are) true for this set